



TRANSTECH 2025

The AIP (Air-Independent Propulsion) model marked a breakthrough in BuildNEO's pursuit of advanced maritime technologies. Utilizing fuel cell technology, it achieves 70–80% energy efficiency and 2–4 weeks of submerged endurance. With metal hydride hydrogen storage, it maximizes space and safety. Its silent, snorkel-free operation ensures superior stealth, a 2500 nm range, and strategic dominance over diesel-electric systems.



IN SEARCH OF
INNOVATION

BuildNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing and social media. By promoting project flexibility and creativity, BuildNEO prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

AIP ENGINE VS DIESEL ENGINE

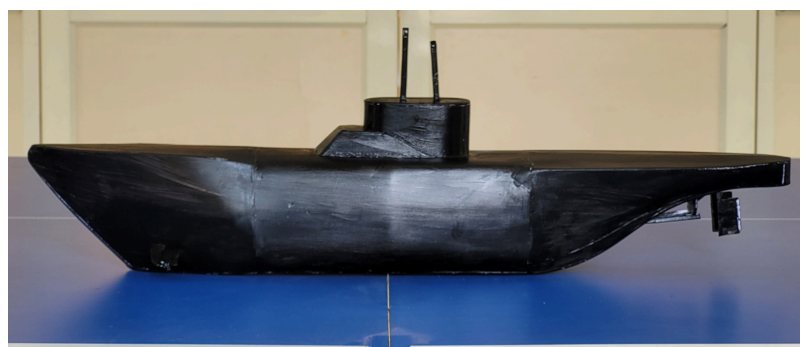
The aim of this project; is to compare Air Independent Propulsion (AIP) engines against the popularly chosen diesel engines in modern day non-nuclear powered submarines. The project consisted of a static submarine and a presentation highlighting the superiority of AIP engines.

OBJECTIVE

- Provide a rational comparison of diesel and AIP engines.
- Analyse the efficiency of AIP-powered submarines.
- Discuss the operational feasibility of AIP engines.
- Enhance engine functioning by usage of metal halides.

SIGNIFICANCE OF AIP

- AIP engines, using fuel cells, achieve 70% energy conversion.
- Range of submerged operation - 2500 nautical miles (approx)
- Submerged speed - 5-8 knots
- Endurance - 2-4 weeks



AIP ENGINE SIGNIFICANCE

The aim of this project; is to explain about the superiority of Air Independent Propulsion (AIP) engines over the popularly chosen diesel engines in modern day non-nuclear powered submarines. The project consisted of a static submarine and a presentation highlighting the significance of AIP engines.

MODEL CHARACTERISTICS

- Static model
- Deck length - 68cm
- Deck width - 8cm
- Height - 11cm
- Base width - 7cm
- Base length - 35cm



OPERATIONAL FEATURES OF AIP

- Range - 1500 to 2500 Nm
 - Endurance - 2 to 4 weeks
 - Near stealth operation
 - Lowest emission rate
 - Best for medium range operation (1000 - 3000 Nm)
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